ZCAS UNIVERSITY

CCS1401 COMPUTING AND ICT SKILLS

FINAL EXAMINATION

WEDNESDAY 27TH MAY, 2026

14:00- 17:00 HRS

TIME ALLOWED: READING AND WRITING – THREE HOURS

UPLOADING ANSWER SCRIPT – 30 MINUTES

INSTRUCTIONS:

1. Section A: Question One in Section A is **compulsory**.
2. Sections B: Answer ***THREE*** questions from this section.
3. This examination paper carries a total of **100 marks**.

SECTION A: All questions are compulsory and must be answered.

Question 1

SmartBite Restaurant plans to introduce an iPad-based ordering system where customers will use iPads placed on tables to browse the menu and place orders electronically. The system will automatically send customer orders to the kitchen and store sales information in a database for analysis.

The restaurant management believes the system will improve customer service, reduce delays, and help management make better business decisions using the collected data.

Required:

a) State and explain ten (10) benefits of introducing the iPad ordering system at the restaurant. (10 Marks)

b) State and explain five (5) possible challenges or issues the restaurant may face when using the system. (10 Marks)

c) Explain how the following Microsoft features can help the restaurant analyse the sales and customer data collected from the system: Use a practical example.

i. Goal Seek (5 marks)

ii. Mail Merge (6 marks)

d) The system stores customer orders in a database linked to other company records. Explain what a database is and state three advantages of storing restaurant orders in a database system. (9 marks)

[Total 40 marks]

SECTION B: Answer any THREE (3) questions.

Question 2

Discuss the following CPU and memory-related units in point form, stating at least five (5) important points for each unit.
Each unit carries 5 marks.

a) RAM (Random Access Memory)

b) ROM (Read Only Memory)

c) Registers

d) Cache Memory

[Total 20 marks]

Question 3

Refer to the 'Sales Data' sheet for all data required to answer the questions.

	A	B	C	D	E	F	G
1	SALES DATA EXTRACT – Q1 2024						
2	*Use this data to answer all questions on the Exam sheet.*						
3							
4	**No.**	**Sales Rep**	**Region**	**Product**	**Units Sold**	**Unit Price ($)**	**Total Sales ($)**
5	1	Alice Banda	North	Laptop	45	1,200.00	
6	2	Brian Mwale	South	Tablet	30	450.00	
7	3	Chanda Phiri	East	Laptop	60	1,200.00	
8	4	Diana Lungu	West	Phone	80	350.00	
9	5	Edward Zulu	North	Tablet	25	450.00	
10	6	Fatima Banda	South	Laptop	55	1,200.00	
11	7	Grace Mwanza	East	Phone	90	350.00	
12	8	Henry Tembo	West	Laptop	40	1,200.00	
13	9	Irene Sakala	North	Phone	70	350.00	
14	10	James Ngoma	South	Tablet	35	450.00	
15	11	Kunda Mutale	East	Laptop	50	1,200.00	
16	12	Lilian Mbewe	West	Tablet	65	450.00	
17	13	Moses Siame	North	Phone	85	350.00	
18							

For the spreadsheet above, write the formulas required to calculate the following.

(Write only the Excel formulas exactly as they would appear in Microsoft Excel using correct Excel syntax. Do not perform the calculations or provide the answers.)

a) Calculate the TOTAL revenue generated by ALL sales representatives combined. (1 marks)

b) Calculate the AVERAGE unit price across all products. (1 marks)

c) Find the HIGHEST total sale value recorded across all 15 transactions.(1 marks)

d) Count how many transactions had Total Sales GREATER THAN $40,000. (3 marks)

e) Calculate total sales for 'Laptop' sold specifically in the 'East' region. (4 marks)

f) For Alice Banda (row 5), write an IF formula that returns 'High Performer' if her Total Sales (G5) exceed $50,000, otherwise return 'Standard'. (4 marks)

g) Write a nested IF using the Unit Price in F5 to classify the product: (6 marks)

- Price > 1,000 → 'Premium'
- Price > 400 → 'Mid-Range'
- Otherwise → 'Budget'

[Total 20 marks]

Question 4

Scenario: A school uses a database called SchoolDB with a table called tblStudents. Study the table data and ALL query screens carefully before answering each part.

Fig. 2 — tblStudents (sample data)

StudentID	FirstName	LastName	Gender	DOB	Mark	Grade	Active
S001	Chanda	Mwale	F	12/03/2008	78	10	Yes
S002	Brian	Banda	M	05/07/2007	45	11	Yes
S003	Miriam	Tembo	F	19/11/2008	62	10	No
S004	James	Phiri	M	30/01/2007	91	11	Yes
S005	Luyando	Banda	F	08/06/2009	55	9	Yes
S006	Peter	Mulenga	M	14/02/2008	83	10	Yes
S007	Alice	Zulu	F	22/09/2007	37	11	No
S008	Moses	Mwale	M	11/12/2009	70	9	Yes

PART A — Reading Query Screens

Each screen below shows a query in Design View (QBE). Read each grid carefully and answer the questions.

Fig. 3 — qryBETWEEN

	FirstName	LastName	Mark	Grade
Sort:			*Descending*	
Show:	☑	☑	☑	☑
Criteria:			*Between 60 And 85*	
Or:				

(a) List ALL students this query will return in the order they would appear. Use Fig. 2 (3 marks)

(b) Would a student with Mark = 85 appear in the results? (2 marks)

Fig. 4 — qryLIKE_Wildcards

	StudentID	FirstName	LastName
Sort:		*Ascending*	
Show:	☑	☑	☑

	StudentID	FirstName	LastName	
Criteria:	*Like "S00#"*		*Like "M*"*	
Or:				

(c) The expression Like "S00#" uses the # wildcard. What does # represent? (3 marks)

(d) Which students from Fig. 2.0 match BOTH criteria in Fig. 4? Name them.

(3 marks)

PART B — Writing Criteria

Write the correct Access criteria expression for each requirement. The field name is shown in [brackets]

(e) [LastName] Students whose surname starts with the letter M and ends with an A

(4 marks)

(f) [FirstName] First name is exactly 5 characters long and ends in "es" — e.g. James.

(5 marks)

Question 5

A computer laboratory at Bank XYZ is experiencing several performance and hardware-related problems. Users have reported slow booting, incorrect system time, frequent system errors, and reduced overall computer performance.

As an ICT technician, you are required to diagnose the possible causes of these problems and suggest appropriate solutions.

Required:

For each problem below, identify the most likely cause and suggest a suitable solution or action.

Instructions

- Answer in table form only (you can use landscape page orientation or two pages).
- Provide clear and correct ICT terms.
- Do not write essays.

	Problem Observed	Likely Hardware Component (1 mark)	Function of the Component (2 marks)	Suggested Solutions (2 mark)

1	Computer shuts down randomly without warning			
2	System date and time resets after shutdown			
3	Computer is slow when running multiple applications			
4	System performance is unstable due to poor communication between hardware components			

[Total 20 Marks]

END OF EXAMINATION